

1.6 Work

Motivation:

Suppose an object moves in a straight line and a constant force F acts on the object in the direction in which the object moves.

In order to compute work done on the object we need the two following formulas:

- (1) $F = m \cdot a$ (*Force = mass $\cdot$ acceleration*, where $a = g = 9.8 \text{ m/s}^2$)
(2) $W = F \cdot d$ (*Work = Force $\cdot$ distance*)

Note: We typically measure

- F in
- d in
- W in

Note: When force is measured in pounds and distance is measure in feet, the unit for work is $\text{lb} \cdot \text{ft}$.

Note: In calculus III work is studied for a much more general (and some would argue much more useful) perspective with vectors.

The main question that will be answered in this section is: How do we calculate the amount of work done when the force acts on the object in the direction in which the object moves, but the force is NOT constant?

Exploration:

Suppose an object moves along a number line from a to b . Suppose that $f(x)$ is a continuous function that represents the force acting on the object at any given point $x \in [a, b]$.

Definition: The work done in moving an object in a straight line from a to b is:

where f is defined just as it was in the exploration.

Example 1: A 200-lb cable is 100ft long and hangs vertically from the top of a tall building. How much work is required to lift the cable to the top of the building?

Question: What if a 500lb trough of water is attached to the end of the cable and is also being lifted?

Example 2: A 10ft chain weighs 25lbs and hangs from a ceiling. Find the work done in lifting the lower end of the chain to the ceiling so that it's level with the upper end.

Example 3: A rectangular tank 2m long, 3m wide, and 1m deep is full of oil that has a density of $900\text{kg}/\text{m}^3$. Find the work required to pump all of the oil out over the top side of the tank.

Example 4: A tank has the shape of an inverted circular cone with height 10ft and base radius 4ft . It is filled with water to a height of 8ft . Find the work required to empty the tank by pumping all of the water to the top of the tank. (The density of the water is $62.4\text{lbs}/\text{ft}^3$)